# ECE 447

## Lesson 31

## Probability, Part 2



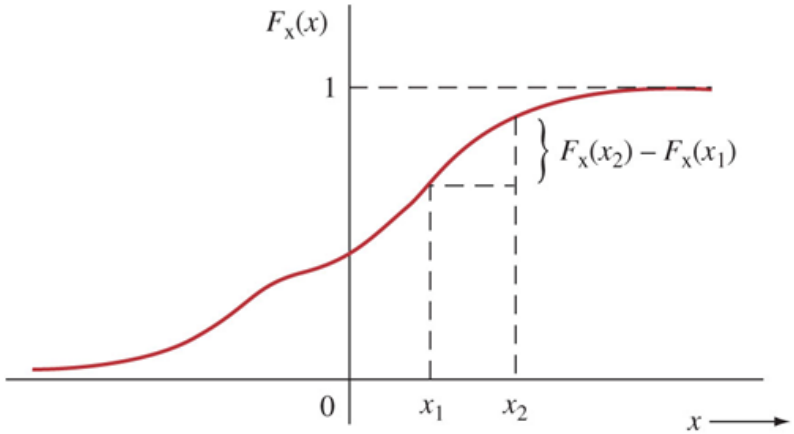
# Schedule and Admin

- Schedule
- Admin
  - **Lab 6**. PDF due tonight (Lesson 31, 4 Nov) to Gradescope.
  - **HW7**. Due Lesson 32 (Fri, 6 Nov) to Gradescope.
  - **HW8**. Assigned today. Due Lesson 34 (Fri, 13 Nov) to Gradescope.
  - **GR2** is Lesson 35 (Tue, 17 Nov), covering Lessons 18-32 (Chapters 5-8). Review day is Lesson 34 (Fri, 13 Nov).

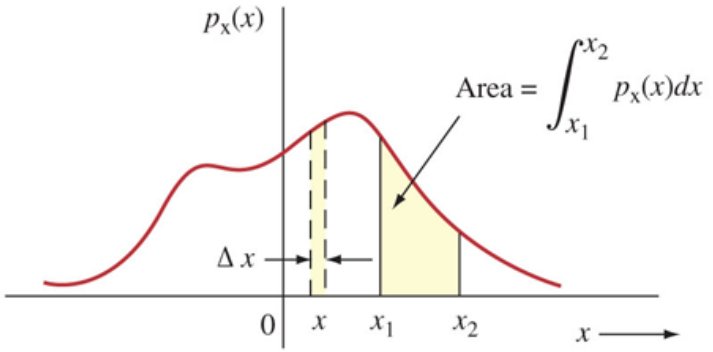
# CDF AND PDF

- CDF: $F_x(x) = P(x \leqslant x)$
  - $F_x(x) \geqslant 0$
  - $F_x(\infty) = 1$
  - $F_x(-\infty) = 0$
  - $F_x(x_1) \leqslant F_x(x_2)$ for $x_1 \leqslant x_2$

- PDF: $p_x(x) = \frac{dF_x(x)}{dx}$
  - $\int_{-\infty}^{\infty} p_x(x)dx = 1$
  - $P(x_1 < x \leqslant x_2) = \int_{x_1}^{x_2} p_x(x)dx$



(a)

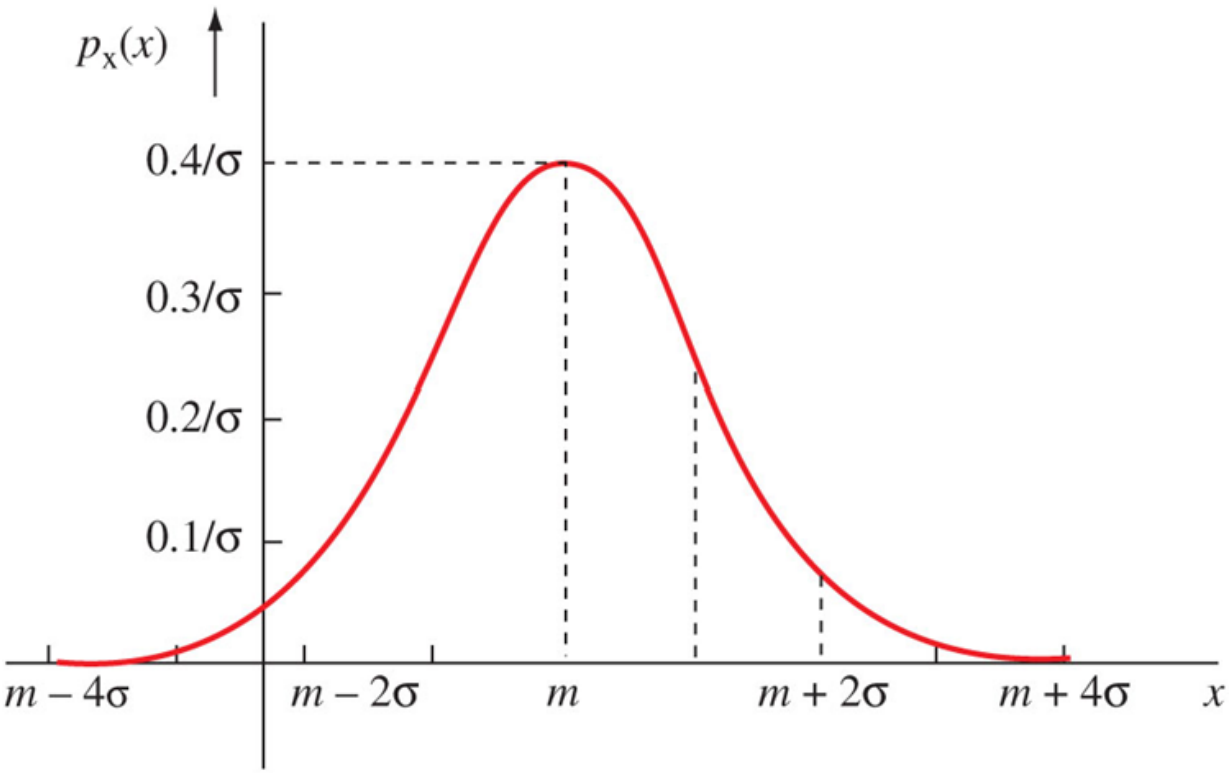


(b)

# Gaussian RV

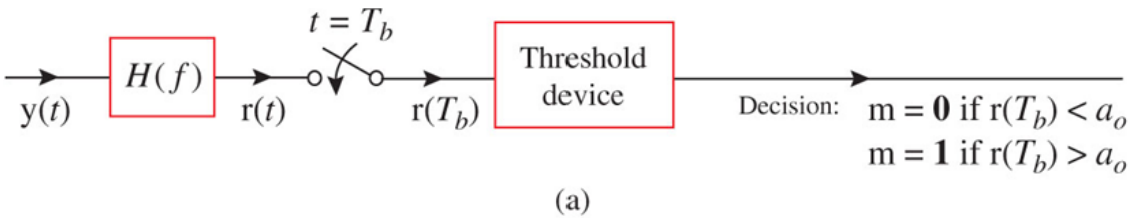
- Standard Gaussian has zero mean and unit variance, $x \sim \mathcal{N}(0,1)$
- General equation for $x \sim \mathcal{N}(m,\sigma)$ is $p_x(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-(x-m)^2/(2\sigma^2)}$
- Typically evaluate using $Q(\cdot)$ functions - area under curve tail

$$P(x > x) = Q\left(\frac{x-m}{\sigma}\right)$$

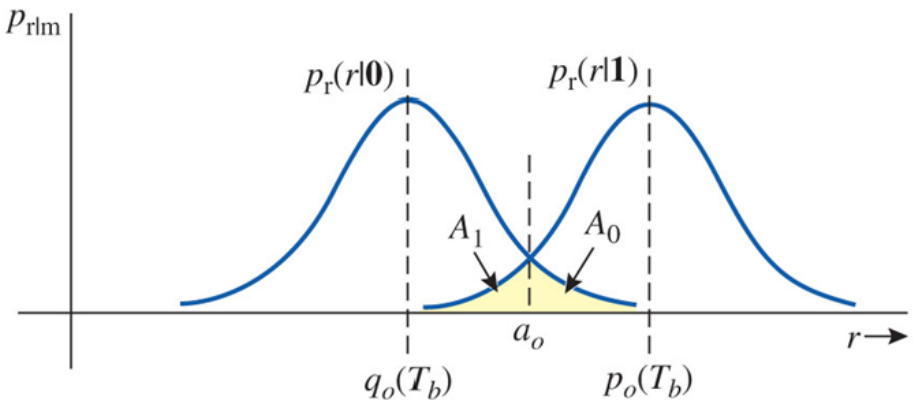


# Gaussian RV in Communication Systems

- Recall (Lsn 25): only care about sample value (test statistic) at the receiver
- $r(T_b) = s_k(T_b) + n(T_b) \rightarrow s_k(T_b)$ deterministic, $n(T_b) \sim \mathcal{N}(0, \sigma)$
- How is $r(T_b)$ distributed? We need to know how to find the *expected value*, $\mathbb{E}[r(T_b)]$
- We'll come back to this problem...




(a)




(b)

# Expected Value

- Aka average value aka mean value
- **For discrete RV:** $\bar{\mathrm{x}} = \mathbb{E}[\mathrm{x}] = \sum_i x_i P_{\mathrm{x}}(x_i)$
- **For continuous RV:** $\bar{\mathrm{x}} = \mathbb{E}[\mathrm{x}] = \int_{-\infty}^{\infty} x p_{\mathrm{x}}(x) dx$
- **Mean of a sum:** mean of the sum is equal to the sum of the means, $\overline{x + y} = \bar{x} + \bar{y}$
- **Mean of a product:** messy, unless independent - then $\overline{xy} = \bar{x} \cdot \bar{y}$

# Moments

- The $n$th moment defined as the mean value of $\mathrm{x}^n$
- **First moment:** expected value, $\mathbb{E}[\mathrm{x}] = \int_{-\infty}^{\infty} x p_{\mathrm{x}}(x) dx$
- **Central moments:** subtract out the mean first, $\mathbb{E}^n[(\mathrm{x} - \overline{\mathrm{x}})^n] = \int_{-\infty}^{\infty} (x - \overline{\mathrm{x}})^n p_{\mathrm{x}}(x) dx$
- **Second central moment:** variance, mean square value minus square of the mean:

$$\sigma_{\mathrm{x}}^2 = \mathbb{E}^2[(\mathrm{x} - \overline{\mathrm{x}})^2] = \int_{-\infty}^{\infty} (x - \overline{\mathrm{x}})^2 p_{\mathrm{x}}(x) dx = \overline{\mathrm{x}^2} - \overline{\mathrm{x}}^2$$

- Variance of a sum of independent RVs is equal to the sum of their variances
  - If $x$ and $y$ are independent RVs and $z = x + y$, then $\sigma_z^2 = \sigma_x^2 + \sigma_y^2$

# Practice: $Q$ Function and Moments

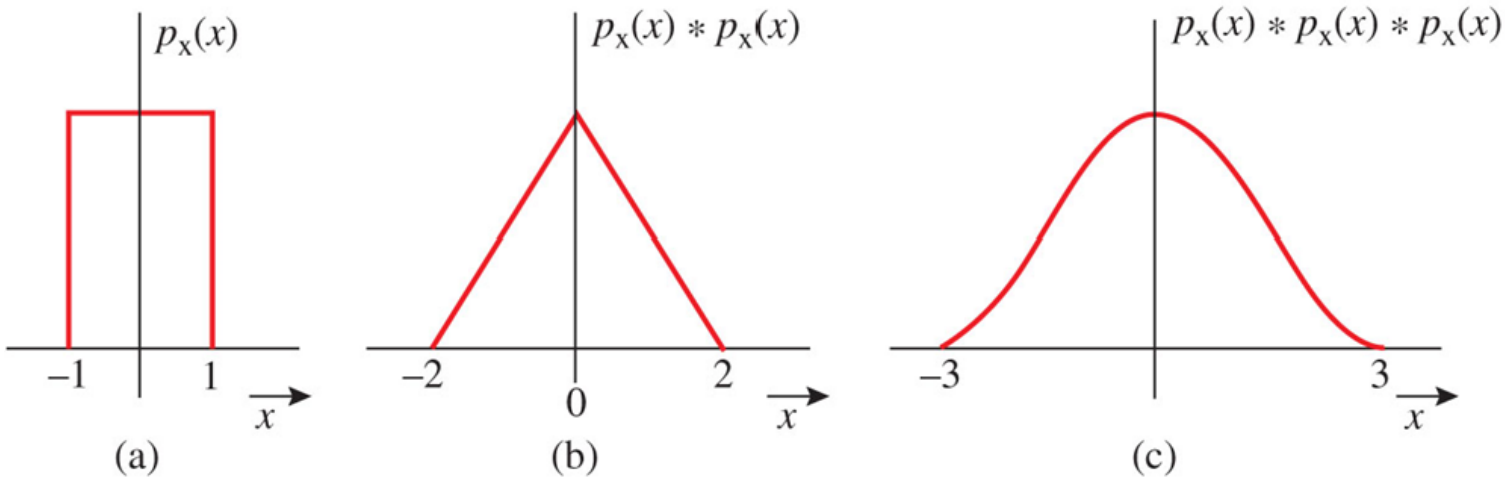
- **(a) Reading a Gaussian.** Detector noise $\mathrm{n} \sim \mathcal{N}(0, \sigma)$ with $\sigma = 0.5$ V, threshold at 0.9 V.
  - $P(\mathrm{n} > 0.9) = Q\left(\frac{x-m}{\sigma}\right) = Q(____) = _____$
  - Now the noise picks up a 0.2 V DC offset. Up or down, and by how much?
  - $P(|\mathrm{n}| > 0.9) = _____$ — why is that not the same as $P(\mathrm{n} > 0.9)$?
  - Handed $p_{\mathrm{x}}(x) = \frac{1}{C\sqrt{2\pi}} e^{-(x-1)^2/8}$, read off $m = ____$ $\sigma = ____$ $C = ____$

- **(b) Moments.** Quantization error is uniform on $\left[-\frac{\Delta}{2}, \frac{\Delta}{2}\right]$, so $p_{\mathrm{x}}(x) = 1/\Delta$ there.
  - $\mathbb{E}[\mathrm{x}] = ____$ by inspection — why? $\quad \mathbb{E}[\mathrm{x}^2] = \int_{-\Delta/2}^{\Delta/2} \frac{x^2}{\Delta} dx = _____$
  - $\sigma_{\mathrm{x}}^2 = \overline{\mathrm{x}^2} - \bar{\mathrm{x}}^2 = _____$ — the PCM quantization noise power from Lesson 19

- HW8 leans on both halves; the mean-square step returns in Lesson 36 as noise power.

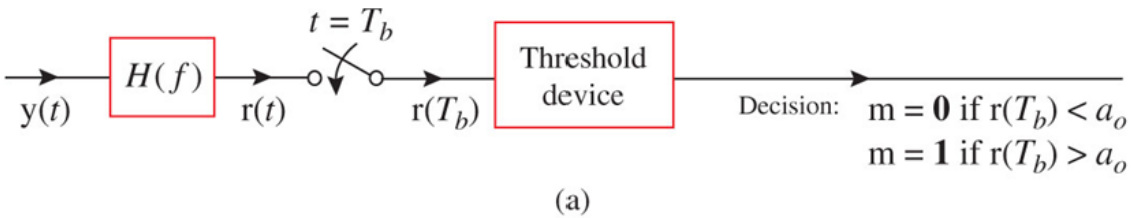
# Central Limit Theorem

- The sum of a large number $n$ independent RVs with mean $\mu$ and variance $\sigma^2$ tends to be a Gaussian RV, regardless of the PDFs of the variables added
- Gaussian mean is $n\mu$
- Gaussian variance is $\sigma^2/n$
- Example 7.25 (similar to future HW problem, 7.7-1)
  - Note the problem defines a discrete RV, $\mathrm{x}_i$
  - Use $\overline{\mathrm{x}} = \mathbb{E}[\mathrm{x}] = \sum_i x_i P_{\mathrm{x}}(x_i)$

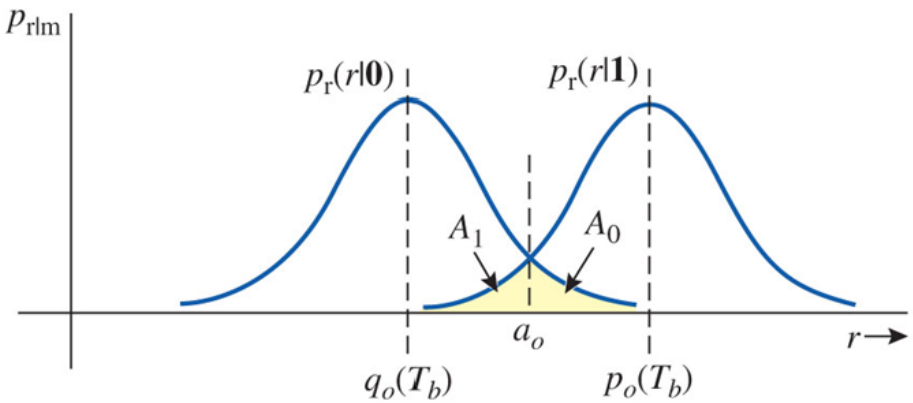


# Gaussian RV in Communication Systems

- Back to this problem!
- $r(T_b) = s_k(T_b) + n(T_b) \rightarrow s_k(T_b)$ deterministic, $n(T_b) \sim \mathcal{N}(0, \sigma)$
- $\mathbb{E}[r(T_b)] = \mathbb{E}[s_k(T_b) + n(T_b)] =$
- $\sigma_r^2 = \mathbb{E}[r^2(T_b)] - \mathbb{E}[r(T_b)]^2 =$




(a)




(b)